STRUCTURAL ANALYSIS / TECHNICAL REPORT

Simply supported beam with uniform load

L = 5 m, q = 10 kN/m downward.

Analysis id: LC1

Source: examples/beams/simple-supported-beam.json

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01 Model overview

2	1	2	1	0	1
Nodes	Members	Supports	Load cases	Combinations	Loads

GEOMETRY & DEFORMATION

Grey: original geometry / Teal: auto-amplified deformation

02 Executive summary

Nodal max |Uy| [m]
0.000000

Max |V| [kN]
25.000000

Max |M| [kNm]
31.250000

Metric	Entity / member	Pos [0-1]	x [m]	Value	Unit
Max |Ux|	A			0.000000	m
Max |Uy|	A			0.000000	m
Max |Rz|	A			0.012401	rad
Max |Fx|	SA			0.000000	kN
Max |Fy|	SA			25.000000	kN
Max |Mz|	SA			0.000000	kNm
Max |N|	M1	0.000000	0.000000	0.000000	kN
Max |V|	M1	0.000000	0.000000	25.000000	kN
Max |M|	M1	0.500000	2.500000	31.250000	kNm

03 Assumptions and limitations

First-order linear elastic 2D analysis. Units: m, rad, kN and kNm. Summary values are absolute magnitudes; detail tables retain their signs.

This is a technical computation report. It does not replace engineering judgement or project-specific normative checks.

04 Nodal displacements

Node	Ux [m]	Uy [m]	Rz [rad]
A	0.000000	0.000000	-0.012401
B	0.000000	0.000000	0.012401

05 Support reactions

Support	Node	Fx [kN]	Fy [kN]	Mz [kNm]
SA	A	0.000000	25.000000	0.000000
SB	B	0.000000	25.000000	0.000000

06 Member end forces

Member	End	N [kN]	V [kN]	M [kNm]
M1	Start	0.000000	25.000000	0.000000
M1	End	0.000000	25.000000	0.000000

07 Internal-force samples

Member	Pos [0-1]	x [m]	N [kN]	V [kN]	M [kNm]
M1	0.000000	0.000000	-0.000000	25.000000	0.000000
M1	0.050000	0.250000	-0.000000	22.500000	5.937500
M1	0.100000	0.500000	-0.000000	20.000000	11.250000
M1	0.150000	0.750000	-0.000000	17.500000	15.937500
M1	0.200000	1.000000	-0.000000	15.000000	20.000000
M1	0.250000	1.250000	-0.000000	12.500000	23.437500
M1	0.300000	1.500000	-0.000000	10.000000	26.250000
M1	0.350000	1.750000	-0.000000	7.500000	28.437500
M1	0.400000	2.000000	-0.000000	5.000000	30.000000
M1	0.450000	2.250000	-0.000000	2.500000	30.937500
M1	0.500000	2.500000	-0.000000	0.000000	31.250000
M1	0.550000	2.750000	-0.000000	-2.500000	30.937500
M1	0.600000	3.000000	-0.000000	-5.000000	30.000000

Internal-force samples / continued

Member	Pos [0-1]	x [m]	N [kN]	V [kN]	M [kNm]
M1	0.650000	3.250000	-0.000000	-7.500000	28.437500
M1	0.700000	3.500000	-0.000000	-10.000000	26.250000
M1	0.750000	3.750000	-0.000000	-12.500000	23.437500
M1	0.800000	4.000000	-0.000000	-15.000000	20.000000
M1	0.850000	4.250000	-0.000000	-17.500000	15.937500
M1	0.900000	4.500000	-0.000000	-20.000000	11.250000
M1	0.950000	4.750000	-0.000000	-22.500000	5.937500
M1	1.000000	5.000000	-0.000000	-25.000000	0.000000

08 Displacement samples

Member	Pos [0-1]	x [m]	GUx [m]	GUy [m]
M1	0.000000	0.000000	0.000000	0.000000
M1	0.050000	0.250000	0.000000	-0.002945
M1	0.100000	0.500000	0.000000	-0.005580
M1	0.150000	0.750000	0.000000	-0.007906
M1	0.200000	1.000000	0.000000	-0.009921
M1	0.250000	1.250000	0.000000	-0.011626
M1	0.300000	1.500000	0.000000	-0.013021
M1	0.350000	1.750000	0.000000	-0.014106
M1	0.400000	2.000000	0.000000	-0.014881
M1	0.450000	2.250000	0.000000	-0.015346
M1	0.500000	2.500000	0.000000	-0.015501
M1	0.550000	2.750000	0.000000	-0.015346
M1	0.600000	3.000000	0.000000	-0.014881
M1	0.650000	3.250000	0.000000	-0.014106
M1	0.700000	3.500000	0.000000	-0.013021
M1	0.750000	3.750000	0.000000	-0.011626
M1	0.800000	4.000000	0.000000	-0.009921
M1	0.850000	4.250000	0.000000	-0.007906
M1	0.900000	4.500000	0.000000	-0.005580
M1	0.950000	4.750000	0.000000	-0.002945
M1	1.000000	5.000000	0.000000	0.000000